Louise Amber **Welsh**

Astronomer

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Employment

Centre for Extragalactic Astronomy — *Durham, UK*

CEA Postdoctoral Fellow — *Jan. 2025 - PRESENT*

INAF Astronomical Observatory of Trieste — *Trieste, Italy*

Postdoctoral Researcher — *Sep. 2023 - Dec. 2024*

University of Milano Bicocca — *Milan, Italy*

Postdoctoral Researcher — *Sep. 2021 - Sep. 2023*

Education

Centre for Extragalactic Astronomy, Durham University — *Durham, UK*

PhD in Astrophysics — *Oct. 2017 - Sep. 2021*

- Thesis: 'A window to the first stars: An investigation of chemically near-pristine environments'
- Advisor: Prof. Ryan Cooke & Prof. Michele Fumagalli

Lancaster University — *Lancaster, UK*

Master of Physics (MPhys): 1st Class (Hons) — *Oct. 2012 - Jul. 2016*

- Thesis: 'Investigating cold dark matter candidates'
- Advisor: Dr. John McDonald

Awards and Fellowships

2025	**CEA Fellowship**, Awarded to pursue observationally based independent research for three years.	*Durham University*
2021	**Keith Nicholas Prize**, Awarded for outstanding overall performance by a postgraduate student.	*Durham University*
2019	**Associate Fellow of the Higher Education Academy**, through the Durham Excellence in Learning and Teaching Awards scheme.	
2019	**Martin and Beate Block Award**, Awarded to a promising researcher at the Aspen winter meeting 'Into the Starlight'.	*Aspen Centre for Physics*
2016	**Azzedine Hammiche Prize**, Awarded for exceptional fourth year project work.	*Lancaster University*

Talks (16 invited, 52 total)

Most recent:

May. 2026	**Galactic metal cycles**, The enrichment and environment of extremely metal-poor DLAs (invited)	*Ireland*
Mar. 2026	**40 years of DLAs**, The curious case of metal-poor DLAs at cosmic noon	*Japan*
Mar. 2026	**CAR seminar**, The curious case of metal-poor DLAs at cosmic noon (invited)	*UK*
Dec. 2025	**Baryon cycle from reionization**, The curious case of metal-poor DLAs at cosmic noon (invited)	*Chile*
July 2025	**BRIDGCE2025**, The curious case of metal-poor DLAs at cosmic noon	*UK*
May. 2025	**INAF Arcetri**, The curious case of metal-poor DLAs at cosmic noon (invited)	*Italy*
May. 2025	**IGM-galaxy connection in the 1st Gyr**, The clustering of metals at the end of reionization (invited)	*Italy*
May. 2025	**EQUALs**, The carbon isotope ratio of metal-poor DLAs	*Italy*
Jan. 2025	**The enigma of first stars in the era of JWST**, Near-pristine DLAs and their chemistry (invited)	*Italy*

Proposal History as Principal Investigator

2025	**Keck II/KCWI**, 1/2 night, 2025A.	*NOIRLab*
2024	**VLT/XSHOOTER**, 8 hours, P113.	*ESO*
2023	**VLT/ESPRESSO 4-UT**, 1.2 nights, P112.	*ESO*
2023	**VLT/UVES**, 10 hours, P111.	*ESO*
2023	**Keck I/HIRES**, 1 night, 2023A.	*NOIRLab*
2022	**VLT/UVES**, 10 hours, P110.	*ESO*
2022	**VLT/ESPRESSO 1-UT**, 7 hours, P109.	*ESO*
2022	**VLT/ESPRESSO 3-UT**, 8 hours, P109.	*ESO*
2021	**VLT/UVES**, 18 hours, P108.	*ESO*
2021	**Keck I/HIRES**, 1 night, 2021B.	*NOIRLab*
2020	**VLT/ESPRESSO 1-UT**, 9 hours, P105.	*ESO*
2020	**VLT/UVES**, 20 hours, P105.	*ESO*
2019	**WHT/ISIS**, 7 nights, 2019B.	*ING*

Proposal History as Primary Scientific Investigator

2023	**WEAVE/LIFU**, 21 hours, 2023B.	*UK PATT*
2020	**Palomar/Hale**, 7 night, 2020A.	*Caltech*

Teaching

2025 -	**Demonstrator**, Astronomy Research Laboratory (Level 2 course)	*Durham University*
2025 -	**Co-advisor**, Student masters project in physics	*Durham University*
2025 -	**Co-advisor**, Student masters project in physics	*Durham University*
2025	**Tutor**, Level 1 weekly physics tutorials	*Durham University*
2023	**Co-advisor**, Student bachelor thesis in physics	*Milano-Bicocca U.*
2022 - 2023	**Demonstrator**, Laboratory of Data Acquisition (postgraduate course)	*Milano-Bicocca U.*
2021	**Advisor**, Nuffield Research Placement	*Durham University*
2019 - 2021	**Demonstrator**, Level 2: Stars and Galaxies	*Durham University*
2018 - 2020	**Demonstrator**, Level 1: Further Mathematics for Geoscientists	*Durham University*
2018 - 2019	**Demonstrator**, Level 1: Maths toolkit for Scientists	*Durham University*

Memberships and activities

- 2025 - **CEA**, Postdoctoral representative for the centre for extragalactic astronomy group Durham
- 2025 - **GHOSTLy**, Project coordinator GEMINI/GHOST large programme for studies using metal absorption lines
- 2025 - **IAU**, Junior member of the International Astronomical Union
- 2025 **NAM2025**, Lead organiser of NAM session with three parallel blocks
- 2024 - **HRMOS**, ISM Working Group Member for HRMOS instrument.
- 2023 - **EQUALS**, Member of the first ESO/ESPRESSO large programme focused on $z \sim 3 - 4$ quasars
- 2023 - **XQR-30**, Member of VLT/X-SHOOTER large programme focused on $z > 5.7$ quasars
- 2023 - **INAF**, Associate member of INAF - Osservatorio Astronomico di Trieste.
- 2023 - **IFPU**, Affiliated member of the Institute for Fundamental Physics in the Universe
- 2023 - 2025 **WEAVE**, IFU Working Group Member for WEAVE QSO survey.
- 2022 **WMAG 2022**, Organising committee member for the 'What Matters around Galaxies - 2022' conference.
- 2021 - **WEAVE**, Member of the WEAVE-QSO survey.
- 2021 - **Peer reviewer**, Astrophysical Journal, Astronomy & Astrophyics, MNRAS.
- 2021 - 2023 **Astrocoffee**, Organiser of weekly astrocoffee seminars at Milano-Bicocca.
- 2021 - 2023 **INAF**, Associate member of INAF - Osservatorio Astronomico di Brera.
- 2020 - 2021 **OCW social**, Member of committee responsible for organising department social events.
- 2020 **DEX XVI**, LOC member for the '2020 Vision: progress and tensions in astronomy' workshop.
- 2019 **Small Galaxies, Cosmic Questions**, LOC member for the 'Small Galaxies, Cosmic Questions' conference.
- 2018 - 2019 **Journal Club**, Convener of a weekly meeting of postgraduate students at Durham University.

Outreach

Planetarium *North East, UK*

Show provider *2018 - 2020*

Delivered shows on the constellations and planets at events (including multiple science festivals) and local schools using an inflatable planetarium.

Pre-school astronomy *Milan, Italy*

Volunteer *2022 - 2023*

Visited local preschools to develop and play astronomy related games with students.

Moon Palace *North East, UK*

Volunteer *2025 -*

Volunteer on mobile school bus that has been converted to hold a functional telescope and engage visitors with astronomy and the cosmos.

Computing Skills

Programming Python, git, high-performance computing, batch systems, RStudio.

Accepted Publications

First author:

- **L. Welsh**, V. D'Odorico, F. Fontanot, R. Davies, S. E. I. Bosman, G. Cupani, G. Becker, L. Keating, E. Ryan-Weber, M. Bischetti, M. Haehnelt, H. Chen, Y. Zhu, S. Lai, M. Hirschmann, L. Xie, and Y. Qin (2025) "The clustering of C IV and Si IV at the end of reionisation: A perspective from the E-XQR-30 survey", A&A 703, 274
- **L. Welsh**, R. Cooke, M. Fumagalli, M. Pettini, and G. C. Rudie (2024) "A survey of extremely metal-poor gas at cosmic noon: Evidence of elevated [O/Fe]", A&A, 691, A285
- **L. Welsh**, R. Cooke, M. Fumagalli, & M. Pettini (2023) "Towards ultra metal-poor DLAs: linking the chemistry of the most metal-poor DLA to the first stars", MNRAS, 525, 527
- **L. Welsh**, R. Cooke, M. Fumagalli, & M. Pettini (2022) "Oxygen-enhanced EMP DLAs: A signpost of the first stars?", ApJ, 929, 158
- **L. Welsh**, R. Cooke, & M. Fumagalli (2021) "The stochastic enrichment of Population II stars", MNRAS, 500, 5214
- **L. Welsh**, R. Cooke, M. Fumagalli, & M. Pettini (2020) "A bound on the $^{12}C/^{13}C$ ratio in near-pristine gas with ESPRESSO", MNRAS, 494, 1411
- **L. Welsh**, R. Cooke, & M. Fumagalli (2019) "Modelling the chemical enrichment of Population III supernovae: the origin of the metals in near-pristine gas clouds", MNRAS, 487, 3363

Second author:

- T. A. M. Berg, **L. A. Welsh**, R. J. Cooke, L. Christensen, V. D'Odorico, S. L. Ellison, and S. López (2025) "Unmixed metals: variations in the enrichment of z $\sim$ 4 sub-damped Lyman α systems", MNRAS, 541, 2405
- R. Cooke and **L. Welsh** (2025) "A determination of the cosmic microwave background temperature using Galactic molecules", MNRAS, 536, 1980
- R. Cooke, **L. Welsh**, M. Fumagalli, & M. Pettini (2020) "A limit on Planck-scale froth with ESPRESSO", MNRAS, 494, 4884

Co-author:

- R. J. Cooke, J. W. Johnson, P. Noterdaeme, M. Pettini, **L. Welsh**, A. Wong, and C. Peroux (2026) "Towards a measurement of the primordial helium isotope ratio", ApJ, in press, arXiv:2605.00122
- P. Noterdaeme, S. Balashev, T. Berg, S. Cristiani, R. Cuellar, G. Cupani, S. Di Stefano, V. D'Odorico, C. Fian, B. Godard, S. López, D. Milaković, A. Trost, and **L. Welsh** (2026) "One H2 molecule per ten million H-atoms reveals sub-pc scale cold overdensities at z$\sim$4", A&A, 707, 66

- S. Rowlands, R. Davies, E. Ryan-Weber, L. Keating, A. Sebastian, G. Becker, M. Bischetti, H. Chen, F. Davies, V. D'Odorico, P. Gaikwad, S. Gallerani, M. Haehnelt, G. Kulkarni, R. Meyer, **L. Welsh**, and Y. Zhu (2026) "E-XQR-30: Evidence for an Increase in the Ionization State of Metal Absorbers from $z \sim 6$ to $z \sim 2$", MNRAS, 546, 1, 2254
- T. A. M. Berg, V. D'Odorico, E. Boera, G. Calderone, R. Cuellar, G. Cupani, S. Cristiani, S. Di Stefano, A. Grazian, F. Guarneri, V. Iršič, S. Lopez, D. Milakovic, P. Noterdaeme, L. Pasquini, M. Viel, and **L. Welsh** (2025) "From the intergalactic to the interstellar scales – EQUALS: a high-resolution legacy survey of gas in the distant Universe using ESPRESSO", Messenger 195, 23
- D. Ďurovčíková, A.-C. Eilers, R. A. Simcoe, **L. Welsh**, R. A. Meyer, J. Matthee, E. V. Ryan-Weber, M. Yue, H. Katz, S. Satyavolu, G. Becker, F. B. Davies, and E. P. Farina (2025) "An Extremely Metal-poor Lyα Emitter Candidate at z = 6 Revealed through Absorption Spectroscopy", ApJL, 987, L33
- D. Milaković, J. K. Webb, P. Molaro, C.-C. Lee, P. Jethwa, G. Cupani, M. T. Murphy, **L. Welsh**, V. D'Odorico, S. Cristiani, R. Génova Santos, C. J. A. P. Martins, N. J. Nunes, T. M. Schmidt, F. A. Pepe, M. R. Zapatero Osorio, Y. Alibert, J. I. González Hernández, P. Di Marcantonio, E. Palle, R. Rebolo, N. C. Santos, S. G. Sousa, and A. Suárez Mascareño (2024) "Isotopic abundance of carbon in the DLA towards QSO B1331+170", MNRAS, 534, 12
- E. H. Nuñez, C. C. Steidel, E. N. Kirby, G. C. Rudie, N. Z. Prusinski, Y. Chen, Z. Zhuang, A. L. Strom, D. K. Erb, M. Pettini, **L. Welsh**, D. S. N. Rupke, and R. J. Cooke (2024) "KBSS-InCLOSE. I. Design and First Results from the Inner Circumgalactic Medium of QSO Line-of-sight Emitting Galaxies at z $\sim$ 2–3", ApJ, 976, 41
- A. Longobardi, M. Fossati, M. Fumagalli, B. Agarwal, E. Lofthouse, M. Galbiati, R. Dutta, T. A. M. Berg, and **L. A. Welsh** (2023) "Towards an automatic approach to modelling the circumgalactic medium: new tools for mock making and fitting of metal profiles in large surveys", RASTI, 2, 470
- R. J. Cooke, P. Noterdaeme, J. W. Johnson, M. Pettini, **L. Welsh**, C. Peroux, M. T. Murphy, and D. H. Weinberg (2022) "Primordial Helium-3 Redux: The Helium Isotope Ratio of the Orion Nebula", ApJ, 932, 60